WORKED SOLUTION

STEP III 2023, Question 12

Probability and Statistics • Combinatorial Probability

STEP III 2023, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Although the two socks of a pair look alike, the selection is a uniformly random set of $2k$ of the $2n$ *physical* socks, so we label every sock and count sets. Then everything is a matter of building a selection in stages: first choose which pairs appear complete, then choose which other pairs contribute a single sock, then choose which sock of each such pair. Part (iii) is the familiar trick $r\binom{n}{r} = n\binom{n-1}{r-1}$, which turns a weighted sum into a sum of probabilities.

Solution

Label the socks so that all $2n$ are distinct. There are $\binom{2n}{2k}$ equally likely selections, and every probability below is (number of favourable selections)$/\binom{2n}{2k}$.

Part (i)

A selection with no complete pair contains at most one sock from each pair, so its $2k$ socks come from $2k$ different pairs. To build one, choose the $2k$ pairs that are represented, in $\binom{n}{2k}$ ways (possible since $2k \le n$), and then choose one of the two socks from each, in 2^{2k} ways. Each such choice gives a different set, and every set with no pair arises this way, so

$$\mathrm{P}(\text{no pair}) = \frac{\binom{n}{2k}\,2^{2k}}{\binom{2n}{2k}}.$$

Part (ii)

Fix r with $0 \le r \le k$. A selection with exactly r complete pairs consists of

- r pairs taken whole ($2r$ socks), and
- $2k - 2r = 2(k-r)$ further socks, all from different pairs (otherwise there would be

more than r pairs), and none from the r pairs already used.

So we choose the r complete pairs in $\binom{n}{r}$ ways; from the remaining $n-r$ pairs choose the $2(k-r)$ pairs that contribute one sock, in $\binom{n-r}{2(k-r)}$ ways; and pick one sock from each of these, in $2^{2(k-r)}$ ways. Again each set with exactly r pairs is produced exactly once, so

$$\mathrm{P}(X_{n,k}=r)=\frac{\binom{n}{r}\binom{n-r}{2(k-r)}\,2^{2(k-r)}}{\binom{2n}{2k}},$$

as required. (When $r=0$ this is part (i) again.)

Part (iii)

Let $1\le r\le k$. Since

$$r\binom{n}{r}=\frac{r\cdot n!}{r!\,(n-r)!}=n\cdot\frac{(n-1)!}{(r-1)!\,(n-r)!}=n\binom{n-1}{r-1},$$

part (ii) gives

$$r\,\mathrm{P}(X_{n,k}=r)=\frac{n\binom{n-1}{r-1}\binom{n-r}{2(k-r)}\,2^{2(k-r)}}{\binom{2n}{2k}}.$$

Now apply part (ii) with $n-1$ pairs, $2(k-1)$ socks drawn and $r-1$ pairs found. This is allowed, because $2(k-1)\le n-1$ follows from $2k\le n$, and $0\le r-1\le k-1$. Since $(n-1)-(r-1)=n-r$ and $(k-1)-(r-1)=k-r$,

$$\mathrm{P}(X_{n-1,k-1}=r-1)=\frac{\binom{n-1}{r-1}\binom{n-r}{2(k-r)}\,2^{2(k-r)}}{\binom{2n-2}{2k-2}}.$$

The numerators match apart from the factor n, so

$$\begin{aligned}\frac{r\,\mathrm{P}(X_{n,k}=r)}{\mathrm{P}(X_{n-1,k-1}=r-1)}&=\frac{n\binom{2n-2}{2k-2}}{\binom{2n}{2k}}=n\cdot\frac{(2n-2)!}{(2k-2)!\,(2n-2k)!}\cdot\frac{(2k)!\,(2n-2k)!}{(2n)!}\\&=n\cdot\frac{2k(2k-1)}{2n(2n-1)}=\frac{k(2k-1)}{2n-1}.\end{aligned}$$

(Dividing was safe: $n-r\ge 2k-r\ge 2(k-r)$, so every binomial coefficient here is positive.) Multiplying up gives the required result.

Ask yourself How does this identity help with $\mathrm{E}(X_{n,k})$?

The expectation is exactly a sum of the left-hand sides, and the right-hand sides add up to a whole probability distribution. The $r=0$ term of the expectation is zero, so

$$\begin{aligned}\mathrm{E}(X_{n,k})=\sum_{r=1}^{k}r\,\mathrm{P}(X_{n,k}=r)&=\frac{k(2k-1)}{2n-1}\sum_{r=1}^{k}\mathrm{P}(X_{n-1,k-1}=r-1)\\&=\frac{k(2k-1)}{2n-1}\sum_{s=0}^{k-1}\mathrm{P}(X_{n-1,k-1}=s).\end{aligned}$$

The variable $X_{n-1,k-1}$ counts pairs among $2(k-1)$ socks, so it takes only the values $0, 1, \ldots, k-1$, and the last sum is 1. Hence

$$\mathrm{E}(X_{n,k}) = \frac{k(2k-1)}{2n-1}.$$

Checks

- With $k = 1$ (two socks), the answer is $\frac{1}{2n-1}$: whichever sock comes first, exactly one of the remaining $2n-1$ socks is its partner.
- Indicators give the mean directly: pair i is complete with probability $\binom{2n-2}{2k-2}/\binom{2n}{2k} = \frac{2k(2k-1)}{2n(2n-1)}$, and multiplying by n pairs gives $\frac{k(2k-1)}{2n-1}$ again.
- For $n = 6$, $k = 3$: part (i) gives $\frac{64}{924} = \frac{16}{231}$ and the mean is $\frac{15}{11}$. A computer enumeration of all selections for every $n \leq 8$ confirmed the distribution in (ii), the identity in (iii) and the mean.

Common slips

- Treating the socks of a pair as truly identical and counting "colour patterns": those are not equally likely, so the counting must be over labelled socks.
- Forgetting the factor $2^{2(k-r)}$ for which sock of each odd pair is taken.
- In (iii), not noticing that $\sum_s \mathrm{P}(X_{n-1,k-1} = s) = 1$, and trying to evaluate the sum by brute force.

Beyond STEP

The second check is linearity of expectation with indicator variables, which finds a mean without ever knowing the distribution; it works even though the events "pair i is complete" are dependent. The identity in (iii) is a size-biasing relation: weighting the distribution by r gives a shifted copy of a smaller problem, a device that also yields the means of the binomial and hypergeometric distributions. Blitzstein and Hwang, *Introduction to Probability*, uses indicators and story proofs like these throughout.

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